Correction - Distributivité simple

Exercice 1

a. $3(x + 4)$
$= 3 \times x + 3 \times 4$
$= 3x + 12$

b. $5(y - 2)$
$= 5 \times y - 5 \times 2$
$= 5y - 10$

c. $7(2x + 1)$
$= 7 \times 2x + 7 \times 1$
$= 14x + 7$

d. $-2(x + 6)$
$= -2 \times x - 2 \times 6$
$= -2x - 12$

e. $-4(3 - a)$
$= -4 \times 3 + 4 \times a$
$= -12 + 4a$

f. $x(x + 5)$
$= x \times x + x \times 5$
$= x^2 + 5x$

Exercice 2

$A = 2(x + 3) + 5x$
$A = 2 \times x + 2 \times 3 + 5x$
$A = 2x + 6 + 5x$
$A = 7x + 6$

$B = 4(2x - 1) - 3$
$B = 4 \times 2x - 4 \times 1 - 3$
$B = 8x - 4 - 3$
$B = 8x - 7$

$C = 3(x + 2) + 2(x - 5)$
$C = 3 \times x + 3 \times 2 + 2 \times x - 2 \times 5$
$C = 3x + 6 + 2x - 10$
$C = 5x - 4$

$D = 5(2y + 1) - 3(y + 4)$
$D = 5 \times 2y + 5 \times 1 - 3 \times y - 3 \times 4$
$D = 10y + 5 - 3y - 12$
$D = 7y - 7$

$E = -2(3x - 4) + 6x$
$E = -2 \times 3x + 2 \times 4 + 6x$
$E = -6x + 8 + 6x$
$E = 8$

$F = x(x + 2) + 3(x - 1)$
$F = x \times x + x \times 2 + 3 \times x - 3 \times 1$
$F = x^2 + 2x + 3x - 3$
$F = x^2 + 5x - 3$

Exercice 3

$G = 5 + (3x - 2)$
$G = 5 + 3x - 2$
$G = 3x + 3$

$H = 7x - (2x + 4)$
$H = 7x - 2x - 4$
$H = 5x - 4$

$I = 4 - (x - 6) + 3x$
$I = 4 - x + 6 + 3x$
$I = 2x + 10$

Exercice 4

1. Aire = $3(x + 5)$

2. $3(x + 5)$
$= 3 \times x + 3 \times 5$
$= 3x + 15$

3. Pour $x = 2$: $3 \times 2 + 15 = 6 + 15 = 21$
L'aire du rectangle est 21 cm².